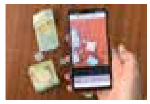
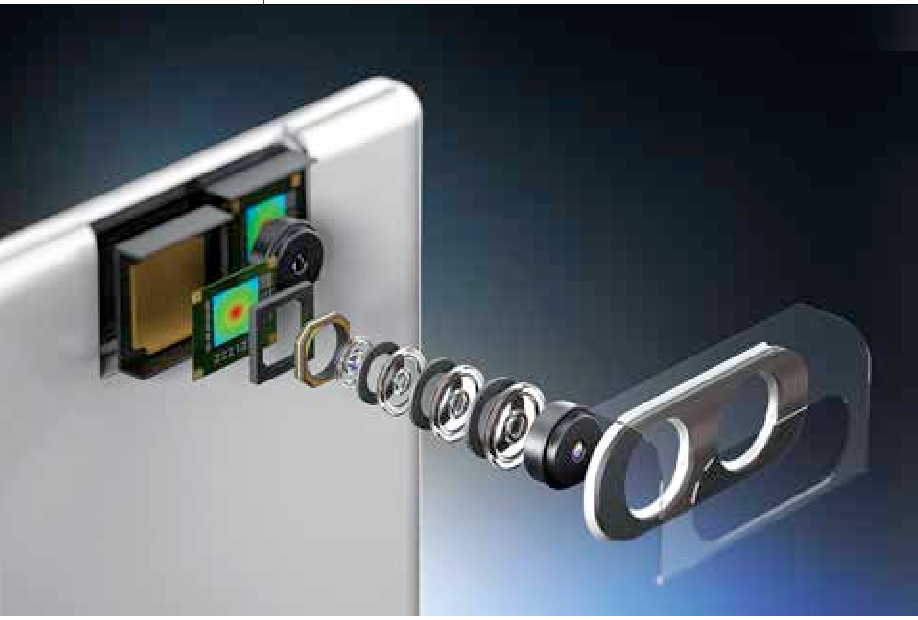
BOE to take on Samsung Display
Samsung Display could lose leadership position when it comes to flexible AMOLED displays to BOE, a Chinese display maker. **https://dgit.in/feb19-75**



App to recognize currency notes
Roshni, an Android app developed by IIT Ropar helps the visually impaired recognise Indian currency notes. **https://dgit.in/feb19-76**



DAPPER SNAPPER

How innovations in smartphone camera hardware are improving photo quality and enabling new capabilities

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There are a few things that consumers need from smartphone cameras. The focusing should be fast, support a wide range of distances, and be able to handle subjects as close to the lens as possible. Good low light performance, HDR capabilities, low shutter lag are priorities. Versatality, in terms of telephoto or wide angle lenses gives users leeway. When it comes to stabilisation, Optical Image Stabilisation (OIS) is preferred over Electronic Image Stabilisation (EIS) to reduce cropping. All of this must work with videos as well.

These are a lot of factors to juggle in very little space, and intense competition means that there is a lot of innovation in smartphone cameras, even though it might not be apparent in iterative releases of models.

ILLUMINATION

The main component of the smartphone cameras is the sensor. Despite the large number of smartphone brands (over a hundred in India alone), there are only a few companies manufacturing the camera sensors for all these brands. Sony is the clear leader, with Samsung coming in the second place. Even some Samsung phones use Sony sensors. Sony and Samsung have nearly identical sensors, so many of the smartphone brands use interchangeable sensors from both companies in the production runs for the same model. Sony sensors use the Exmor brand name, while Samsung calls its sensors ISOCELL.

Considering the low power consumption and speed, most smartphone cameras use CMOS sensors. The pixels are arranged in a matrix, which has layers of components. There is an on chip lens for each pixel, then a filter in one of the RGB colours and a photodiode. There is also a layer of metal wiring and transistors within a silicon substrate. One of the innovations in smartphone camera sensors is back side illumination (BSI). Older (and cheaper) sensors used something called front side illumination (FSI), where the silicon substrate was actually sandwiched between the on chip lens and the photodiode. This meant that some light would be lost because of deflection from the wiring. BSI sensors were expensive, and used in security cameras, astronomical instruments and microscopes, before Sony pioneered a methods for mass manufacturing them. This is the innovation that gave Sony the lead in supplying sensors to smartphone cameras.

SIZE DOES MATTER

There is a delicate balancing act, when it comes to the actual sizes of the pixels, the individual imaging elements on the sensors. Smaller pixel sizes are advantageous as it occupies less real estate on the circuit board and requires smaller and cheaper lenses. On the other hand, larger pixels perform better in low light conditions, generally improve the dynamic range and just produce better images.

The latest sensors by both Samsung and Sony, use pixels that are 0.8 µm in size. Individual pixel sizes reduced beyond the 1 µm limit only in 2017. The smaller size allows for packing in more pixels onto the sensor, and increases the megapixel count. While Sony offers sensors of up to 48 megapixels, Samsung goes up to 32. The latest offering by Sony, with the 48 meg-